

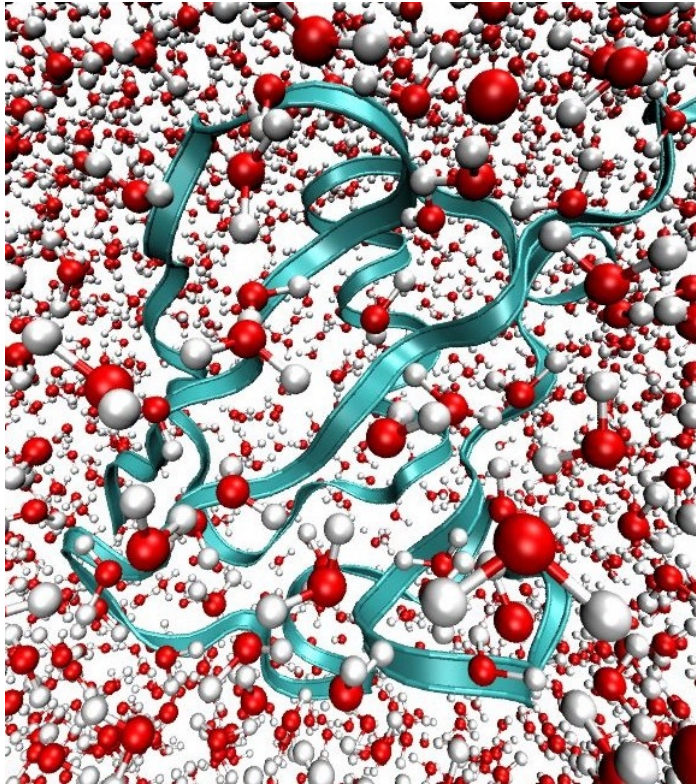
Introduction to simulations with simplified models

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What is a simulation?



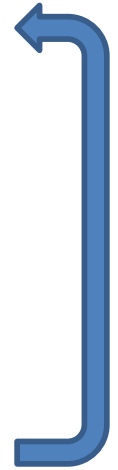
Basic physics

$$F = -\frac{d}{dx}U(x)$$

$$F = M \cdot a$$

$$v(t) = v_0 + a \cdot t$$

$$x(t) = x_0 + v \cdot t$$



- Simulations increment the positions and velocities, using small time steps
 - Many many times....
- One calculates the force between all N^2 atom pairs in the simulated system many times

Objective

- We are going to perform some simulations of a small protein

Software to be used

- smog2 – package that reads a PDB file and then defines a structure-based “SMOG” model
- Python – we have already installed libraries called OpenMM and OpenSMOG, which will be used to perform the simulation
- vmd – “visual molecular dynamics” allows us to visualize the trajectories

Controlling temperature

- Temperature is the parameter that governs the average kinetic energy

$$\left\langle \frac{mv^2}{2} \right\rangle = \frac{3}{2} k_B T$$

- Left side is the average kinetic energy of a single atom.
 k_B is the Boltzmann constant
- T is the temperature
- In a simulation, specifying a temperature is accomplished by ensuring that the average kinetic energy is correct

Let's explore the role of temperature in molecular dynamics

At room temperature, proteins will constantly deform. We would like to inspect those fluctuations.

Caution!

- Commands will be shown in the font `courier`
- DO NOT COPY AND PASTE COMMANDS.
Hidden characters will interfere with the commands. Must type them!

When in the terminal (on the cluster)

Issue the following command to retrieve all workshop files

```
wget https://smog-server.org/workshops/CTBP_2023/smog.tar
```

“untar” the files with:

```
tar xvf smog.tar
```

Now, you should have a directory called “ctbp”

Go into “smog” with :

```
cd smog
```

Prepare a simulation

First, we need to generate a model. For this we will use the “SMOG” software to generate an all-atom structure-based model. However, before we can do that, we need to clean up the PDB file:

```
smog_adjustPDB -i 2CI2.pdb -removewater -o ci2.pdb
```

This takes care of peculiar aspects of PDB files that can confuse simulation software, such as water molecules.

Prepare a simulation

Next, we need to generate a potential energy function.

```
smog2 -i ci2.pdb -AA -dname ci2 -openSMOG
```

“-AA” indicates we want to use the default all-atom model

“-i” indicates that we want to generate a model based on this pdb file

“-dname ci2” indicates that all output files should be named ci2.<suffix>

“-OpenSMOG” indicates that we want to generate files for a specific simulation tool (OpenSMOG)

If smog is successful, you can use “ls” and see several new files

gro (coordinate) file

Command:

```
more ci2.gro
```

```
Gro file for a structure based model, generated with SMOG Version 2.5-beta  
521
```

1ASN	N	1	1.327	-3.751	0.424
1ASN	CA	2	1.191	-3.707	0.460
1ASN	C	3	1.194	-3.556	0.490
1ASN	O	4	1.287	-3.485	0.447
1ASN	CB	5	1.134	-3.791	0.573
1ASN	CG	6	1.201	-3.926	0.588
1ASN	OD1	7	1.137	-4.023	0.632
1ASN	ND2	8	1.328	-3.935	0.552
2LEU	N	9	1.092	-3.513	0.561
2LEU	CA	10	1.076	-3.374	0.604
2LEU	C	11	0.931	-3.337	0.634
2LEU	O	12	0.839	-3.366	0.556
2LEU	CB	13	1.142	-3.283	0.499
2LEU	CG	14	1.228	-3.169	0.551
2LEU	CD1	15	1.312	-3.212	0.671
2LEU	CD2	16	1.319	-3.122	0.437
3LYS	N	17	0.914	-3.277	0.752
3LYS	CA	18	0.784	-3.222	0.793
3LYS	C	19	0.804	-3.070	0.778
3LYS	O	20	0.893	-3.014	0.843

```
--More--(4%)
```

When done looking at file, hit "q"

ci2.top (force field) file

Indicates which atom types
are in this simulation

Name of the complex

Specific atoms in this system

Chemical bonds

bond angles

```
...
[ defaults ]
; nbfunc comb-rule gen-pairs fudgeLJ fudgeQQ
  1       1       no       1       1

[ atomtypes ]
; name  mass      charge  ptype c6          c12
  NB_1   1.0000   0.000000  A    0.00000e+00   5.96046e-09

[ moleculetype ]
; name      nrexcl
  Macromolecule  3

[ atoms ]
;  nr      type  resnr  residue  atom  cgnr  charge
   1      NB_1    1     ASN      N     1
   2      NB_1    1     ASN     CA     2
   3      NB_1    1     ASN     C     3

...
[ bonds ]
; ai  aj      func      r0(nm)      Kb
  1    2      1      1.472425210e-01  1.000000000e+04
  2    3      1      1.532591596e-01  1.000000000e+04

...
[ angles ]
; ai  aj  ak      func      th0(deg)      Ka
  1    2   3      1      1.088732556e+02  8.000000000e+01
  1    2   5      1      1.110327081e+02  8.000000000e+01

...
[ exclusions ]
; ai  aj
  15  202
...
```

ci2.xml (native contact) file

```
<!--      OpenSMOG xml file generated by SMOG2.
; Structure-based "SMOG" model for use with OpenMM
; Generated by the SMOG tool version 2.5-beta
; date: Tue Jun 13 20:59:06 CDT 2023
; hostname: gn03.cluster.local
; SMOG2 called with the following flags: -i ci2.pdb -AA -dname ci2 -openSMOG
-->
<OpenSMOGforces OpenSMOGversion="1.1.2">
  <contacts>
    <contacts_type name="contact_1-6-12">
      <expression expr="A/r^12-B/r^6"/>
        <parameter>A</parameter>
        <parameter>B</parameter>
        <interaction i="15" j="202" A="3.27221e-04" B="2.76186e-02"/>
        <interaction i="15" j="217" A="9.18410e-04" B="4.62699e-02"/>
        <interaction i="15" j="358" A="1.04580e-03" B="4.93746e-02"/>
        <interaction i="16" j="356" A="2.02776e-04" B="2.17415e-02"/>
        <interaction i="16" j="358" A="1.02948e-04" B="1.54913e-02"/>
        <interaction i="17" j="189" A="1.11284e-03" B="5.09326e-02"/>
        <interaction i="20" j="189" A="5.38071e-04" B="3.54160e-02"/>
        <interaction i="20" j="198" A="2.79653e-04" B="2.55323e-02"/>
        <interaction i="20" j="200" A="1.10629e-04" B="1.60588e-02"/>
        <interaction i="20" j="202" A="1.98119e-04" B="2.14903e-02"/>
        <interaction i="20" j="516" A="1.15904e-05" B="5.19792e-03"/>
        <interaction i="21" j="189" A="2.08416e-04" B="2.20417e-02"/>
        <interaction i="21" j="190" A="9.97424e-05" B="1.52483e-02"/>
        <interaction i="23" j="189" A="1.16150e-03" B="5.20343e-02"/>
        <interaction i="23" j="190" A="2.05992e-05" B="6.92955e-03"/>
        <interaction i="23" j="193" A="5.78892e-07" B="1.16166e-03"/>
        <interaction i="24" j="62" A="6.60878e-04" B="3.92501e-02"/>
        <interaction i="25" j="71" A="1.39245e-07" B="5.69731e-04"/>
        <interaction i="25" j="159" A="2.00781e-04" B="2.16342e-02"/>
        <interaction i="25" j="193" A="1.40097e-07" B="5.71471e-04"/>
        <interaction i="27" j="516" A="4.58955e-06" B="3.27088e-03"/>
      </expression>
    </contacts_type>
  </contacts>
</OpenSMOGforces>
```

Simulate the protein

- We will use OpenSMOG/OpenMM, which are python libraries. Before we perform a simulation, let's look at a file we will use: Print the contents to screen with:

```
more run.cool.py
```

- Since we have already prepared the scripts, we can run the simulation with a simple command:

```
python run.cool.py
```

Visualize your simulation

- Retrieve your data (replace **hpcN** with your name):

```
scp -J hpcN@md.rice.edu hpcN@aries.rice.edu:/home/hpcN/smog/ci2.pdb .
```

```
scp -r -J hpcN@md.rice.edu hpcN@aries.rice.edu:/home/hpcN/smog/output.cool .
```

- In VMD: first load ci2.pdb and then load output.cool/ci2.AA_trajectory.dcd
- What does the simulation look like? Does it unfold?

Simulate the protein

- Now, let's heat the system up. Print the settings to screen with:

```
more run.hot.py
```

- Since we have already prepared the scripts, we can run the simulation with a simple command:

```
python run.hot.py
```

- Download output.hot and visualize the trajectory

Simulate the protein

- Now, let's cool it back down up. Print the settings to screen with:

```
more run.quench.py
```

- Since we have already prepared the scripts, we can run the simulation with a simple command:

```
python run.quench.py
```

- Download output.quench and visualize trajectory

Congrats!

- You have just simulated a protein!
- Now, how about the ribosome?

Prepare a ribosome simulation

First, we need to clean up the PDB file:

```
smog_adjustPDB -i ribosome.pdb -o ribo.pdb
```

Next, we need to generate a potential energy function.

```
smog2 -i ribo.pdb -AA -dname ribosome -openSMOG
```

Simulate the ribosome

- Again, we will use OpenSMOG/OpenMM. Let's look at a file we will use: Print the contents to screen with:

```
more run.ribosome.py
```

- Since we have already prepared the scripts, we can run the simulation with a simple command:

```
python run.ribosome.py
```

Now, copy the data back to your machine and visualize!!

Next step

- Join my research group, you're a pro!